EXP NO: 14 Experiment to understand feature scaling.

AIM:

To understand and implement feature scaling techniques such as Normalization and Standardization on a given dataset using Python.

PROGRAM:

```
import numpy as np

import pandas as pd

from google.colab import files


uploaded = files.upload()

df = pd.read_csv("pre-process_datasample.csv")

df


df.head()


df.info()


df["Country"] = df["Country"].fillna(df["Country"].mode()[0])


features = df.iloc[:, :-1].values

label = df.iloc[:, -1].values


from sklearn.impute import SimpleImputer


age = SimpleImputer(strategy="mean", missing_values=np.nan)

salary = SimpleImputer(strategy="mean", missing_values=np.nan)


age.fit(features[:, [1]])
```

```
salary.fit(features[:, [2]])

features[:, [1]] = age.transform(features[:, [1]])
features[:, [2]] = salary.transform(features[:, [2]])

features

from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, [0]])

Country

final_set = np.concatenate((Country, features[:, [1, 2]]), axis=1)
final_set

from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
sc.fit(final_set)

feat_standard_scaler = sc.transform(final_set)
feat_standard_scaler

from sklearn.preprocessing import MinMaxScaler

mms = MinMaxScaler(feature_range=(0, 1))
```

mms.fit(final_set)

feat_minmax_scaler = mms.transform(final_set)

feat_minmax_scaler

OUTPUT:

```
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [1.        , 0.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

RESULT:

Thus, feature scaling techniques were successfully implemented on the given dataset to bring numerical features to a common scale.